

```
docker run -p 8080:8080 -p 50000:50000 -d -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
```

Docker run options:



Expose port 8080	➔	By default runs on that port
Expose port 50000	➔	Master / Slave Communication
Run in detached mode	➔	run container in bakground
Bind named volume	➔	persist data of Jenkins

```
C:\Tests\Docker\Tutorial-Jenkins-01>docker run -p 8080:8080 -p 50000:50000 -d -v jenkins_home:/var/jenkins_home jenkins/jenkins:jdk21
08a28b9fcbf3606e862a0091e914dbfe13f3ac23afde83bcd3c32aba104f221

C:\Tests\Docker\Tutorial-Jenkins-01>docker logs 08a28
Running from: /usr/share/jenkins/war
webroot: /var/jenkins_home/war
2025-10-29 11:17:26.060+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2025-10-29 11:17:30.952+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2025-10-29 11:17:31.139+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.3; built: 2025-10-16T14:10:03.576Z; git: f8d520dade4123ea8ce53fb1b9a
c3a7a936a9d0f; jvm 21.0.8+9-LTS
2025-10-29 11:17:33.828+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJs
pServlet
2025-10-29 11:17:34.423+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2025-10-29 11:17:36.866+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /var/jenkins_home found at: EnvVars.masterEnvVars.get("JEN
KINS_HOME")
2025-10-29 11:17:37.544+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@1744a475{Jenkins v2.534/,b=fil
e:///var/jenkins_home/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@444cc791{STARTED}}
2025-10-29 11:17:37.674+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@173797f0(HTTP/1.1, (http/1.1)){0.0.0.0:8080}
2025-10-29 11:17:37.762+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@2c0b4c83(STARTING)[12.1.3,sto=0] @14540ms
2025-10-29 11:17:37.766+0000 [id=34] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2025-10-29 11:17:38.862+0000 [id=33] INFO jenkins.model.Jenkins<init>: Starting version 2.534
2025-10-29 11:17:39.512+0000 [id=42] INFO jenkins.InitReactorRunner$1onAttained: Started initialization
2025-10-29 11:17:39.557+0000 [id=44] INFO jenkins.InitReactorRunner$1onAttained: Listed all plugins
2025-10-29 11:17:44.010+0000 [id=44] INFO jenkins.InitReactorRunner$1onAttained: Prepared all plugins
2025-10-29 11:17:44.026+0000 [id=43] INFO jenkins.InitReactorRunner$1onAttained: Started all plugins
2025-10-29 11:17:44.031+0000 [id=47] INFO jenkins.InitReactorRunner$1onAttained: Augmented all extensions
2025-10-29 11:17:44.541+0000 [id=46] INFO jenkins.InitReactorRunner$1onAttained: System config loaded
2025-10-29 11:17:44.544+0000 [id=46] INFO jenkins.InitReactorRunner$1onAttained: System config adapted
2025-10-29 11:17:44.547+0000 [id=46] INFO jenkins.InitReactorRunner$1onAttained: Loaded all jobs
2025-10-29 11:17:44.557+0000 [id=43] INFO jenkins.InitReactorRunner$1onAttained: Configuration for all jobs updated
2025-10-29 11:17:44.902+0000 [id=61] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2025-10-29 11:17:46.382+0000 [id=43] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> a18f2cc8f20c4c7cb991f684f83420b5
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
```

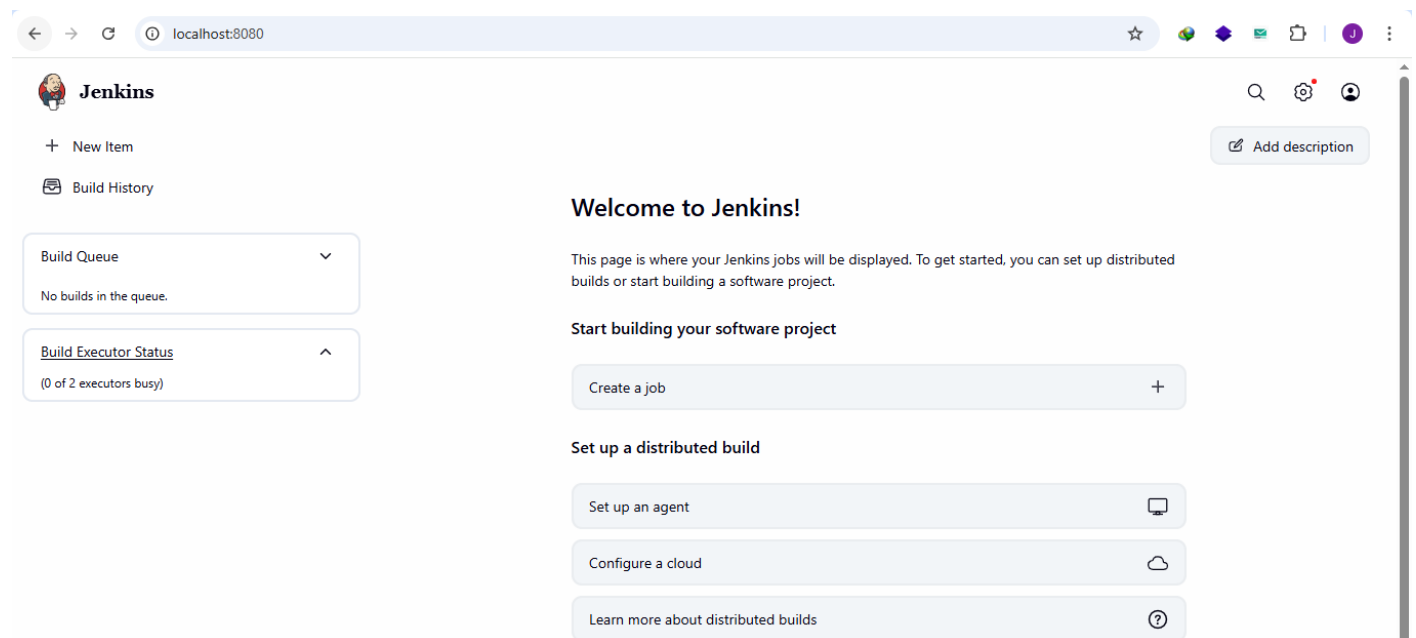
Activate Windows
Go to Settings to activate Windows.

Then we can go to: **localhost:8080**

Then we choose to install the suggested plugins

Then we create new admin user

Then we start Jenkins:



Types of Jenkins Projects

Freestyle

simple, single tasks

e.g. run tests

Pipeline

whole delivery cycle

e.g. test | build | ..

for a **single branch**

Multibranch Pipeline

like pipeline

for **multiple branches**

localhost:8080/manage/

Jenkins / Manage Jenkins

Manage Jenkins

Search settings

Building on the built-in node can be a security issue. You should set up distributed builds. See [the documentation](#).

Set up agent Set up cloud Dismiss

System Configuration

- System**
Configure global settings and paths.
- Tools**
Configure tools, their locations and automatic installers.
- Plugins**
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Nodes**
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**
Add, remove, and configure cloud instances to provision agents on-demand.
- Appearance**
Configure the look and feel of Jenkins.

Security

- Security**
Secure Jenkins; define who is allowed to access/use the system.
- Credentials**
Configure credentials.
- Credential Providers**
Configure the credential providers and types.

localhost:8080/manage/pluginManager/available

Jenkins / Manage Jenkins / Plugins

Plugins

Search available plugins

Install Available plugins Installed plugins Advanced settings Download progress

Install	Name	Released	Health
☐	Pipeline Graph Analysis 245.v88f03631a_b_21 Library plugins (for use by other plugins) Provides a REST API to access pipeline and pipeline run data.	1 mo 19 days ago	97
☐	PAM Authentication 1.12 Security Adds Unix Pluggable Authentication Module (PAM) support to Jenkins	7 mo 28 days ago	95
☐	JavaMail API 1.6.2-11 Library plugins (for use by other plugins) This plugin provides the JavaMail API for other plugins.	8 mo 8 days ago	96
☐	Command Agent Launcher 123.v37cfdc92ef67 Agent Management Allows agents to be launched using a specified command.	6 mo 21 days ago	100
☐	Oracle Java SE Development Kit Installer 83.v417146707a_3d Allows the Oracle Java SE Development Kit (JDK) to be installed via download from Oracle's website.	9 mo 10 days ago	100

Activate Windows
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Branch Source = Project you want to build

To create *multibranch pipeline* we create new item with name and select *multibranch pipeline*:

localhost:8080/job/my-pipeline/configure

Jenkins / my-pipeline / Configuration

Configuration

General Branch Sources Build Configuration Scan Multibranch Pipeline Triggers Orphaned Item Strategy Appearance Health metrics Properties

General

Enabled

Display Name ?

Description

Plain text Preview

Branch Sources

Add source

Save Apply

Activate Windows
Go to Settings to activate Windows.

For branches we select by name with Regular expression, where RE from java

localhost:8080/job/my-pipeline/configure

Jenkins / my-pipeline / Configuration

Configuration

- General
- Branch Sources**
- Build Configuration
- Scan Multibranch Pipeline Triggers
- Orphaned Item Strategy
- Appearance
- Health metrics
- Properties

Branch Sources

Git

Project Repository ?
https://github.com/gaaferHajji2/j-l-jenkins-test.git

Credentials ?
- none - + Add

Behaviors

Discover branches ?

Filter by name (with regular expression)

Regular expression ?
.|

Save Apply

Visual Studio Code

Activate Windows
Go to Settings to activate Windows.

To add new credentials for Jenkins for our pipeline:

- From settings we select credentials and select system, then we select global credentials, then we add new ones:

The first screenshot shows the Jenkins 'Credentials' page. Under 'Stores scoped to Jenkins', the 'System' store is selected (indicated by a red arrow). The second screenshot shows the 'System' store page, where the 'Global credentials (unrestricted)' domain is selected (indicated by a red arrow). The third screenshot shows the 'Global credentials (unrestricted)' page, where the '+ Add Credentials' button is highlighted (indicated by a red arrow).

The system will not be available for pipelines

Credential Scopes

System	→	Only available on Jenkins server NOT for Jenkins jobs
Global	→	Everywhere accessible

Tabweer Web: x Self Service x Roro Harb x Issues - ITCP x Qwen x Deployment x New creden x gaferHaji2 x + -

localhost:8080/manage/credentials/store/system/domain/_/newCredentials

Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr...

New credentials

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Global (Jenkins, nodes, items, all child items, etc)

System (Jenkins and nodes only)

Blank username: did you mean to use secret text credentials instead?

☐ Treat username as secret ?

Password ?

Create

Activate Windows
Go to Settings to activate Windows.

Type here to search 28°C مشمس 1:23 PM 11/3/2025



Tabweer Web: x Self Service x Roro Harb x Issues - ITCP x Qwen x Deployment x New creden x gaferHaji2 x + -

localhost:8080/manage/credentials/store/system/domain/_/newCredentials

Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr...

New credentials

Kind

Username with password

Username with password

GitHub App

SSH Username with private key

Secret file

Secret text

Certificate

Blank username: did you mean to use secret text credentials instead?

☐ Treat username as secret ?

Password ?

Create

Activate Windows
Go to Settings to activate Windows.

Type here to search 28°C مشمس 1:23 PM 11/3/2025

ID = Reference for your credentials

Credential Scopes

Project → Limited to project, **ONLY** with multibranch pipeline

Pipeline Syntax

Scripted

- first syntax
- Groovy engine
- advanced scripting capabilities, high flexibility
- difficult to start

```
dev / Jenkinsfile
1 pipeline {
2
3     agent any
4
5     stages {
6
7         stage("build") {
8
9             steps {
10
11             }
12
13         }
14
15     }
16 }
```

Declarative

- recent addition
- easier to get started, but not that powerful
- pre-defined structure

Note: pipeline and agent in declarative way are the similar to node {} in groovy script

Required Fields of Jenkinsfile

```
dev / Jenkinsfile
1 pipeline {
2
3   agent any
4
5   stages {
6
7     stage("build") {
8
9       steps {
10
11       }
12     }
13   }
14 }
15
16 node {
17   // groovy script
18 }
```

- "pipeline" must be top-level
- "agent" - where to execute
- "stages" - where the "work" happens
 - "stage" and "steps"

Example about basic Jenkinsfile:

```
pipeline{  
  agent any  
  stages {  
    stage("build"){  
      steps {  
        echo 'building the application...'  
      }  
    }  
    stage("test"){  
      steps {  
        echo 'testing the application...'  
      }  
    }  
    stage("deploy"){  
      steps {  
        echo 'deploying the application...'  
      }  
    }  
  }  
}
```
